



Lecture notes for Applies statistics, Probability theory

Lecture 11.

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Presentation Overview

ANOVA -
Analysis of
variance

Matlab and
ANOVA

ANOVA - Analysis of variance

One-Way ANOVA

Two-Way ANOVA

Matlab and ANOVA

One-Way ANOVA

Two-Way ANOVA



Section Overview

ANOVA -
Analysis of
variance

One-Way ANOVA

Two-Way ANOVA

Matlab and
ANOVA

ANOVA - Analysis of variance

One-Way ANOVA

Two-Way ANOVA



Concept question

ANOVA -
Analysis of
variance

One-Way ANOVA
Two-Way ANOVA

Matlab and
ANOVA

A dog show provides an example. A dog show is not a random sampling of the breed: it is typically limited to dogs that are adult, pure-bred, and exemplary. A histogram of dog weights from a show is likely to be rather complicated. Suppose we wanted to predict the weight of a dog based on a certain set of characteristics of each dog.

One way to do that is to explain the distribution of weights by dividing the dog population into groups based on those characteristics. A successful grouping will split dogs such that (a) each group has a low variance of dog weights (meaning the group is relatively homogeneous) and (b) the mean of each group is distinct (if two groups have the same mean, then it isn't reasonable to conclude that the groups are, in fact, separate in any meaningful way).

Source: Wikipedia

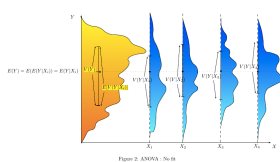


Concept question

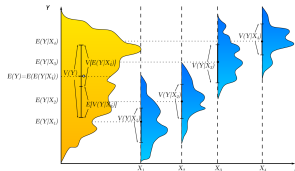
ANOVA - Analysis of variance

One-Way ANOVA
Two-Way ANOVA

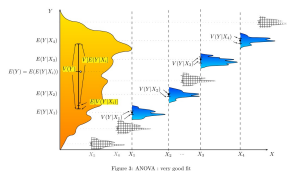
Matlab and
ANOVA



(a) No fit



(b) Fair fit



(c) Very good fit

Fig. 1: Result of different grouping methods

- (a) No fit: Young vs old, and short-haired vs long-haired
- (b) Fair fit: Pet vs Working breed and less athletic vs more athletic
- (c) Very good fit: Weight by breed



ANOVA

ANOVA - Analysis of variance

One-Way ANOVA

Two-Way ANOVA

Matlab and
ANOVA

The ANOVA deals with problems, where the value of the inspected random variable is not only dependent from the randomness, but also from one or more systematic or random factors.

ANOVA, or Analysis of Variance, is a test used to determine differences between research results from three or more unrelated samples or groups.



ANOVA

ANOVA -
Analysis of
variance

One-Way ANOVA

Two-Way ANOVA

Matlab and
ANOVA

Suppose that we have k populations (or groups) that we are studying, and we take an independent random sample from each one. The statistical model that forms the basis for ANOVA is given as

$$x_{ij} = \mu_i + \epsilon_{ij},$$

where x_{ij} is the j -th sample unit in the i -th group. The mean of the i -th group is μ_i . The error terms ϵ_{ij} are independent normally distributed random variables with a mean of zero and variance σ^2 . The indexes take on values $i = 1, \dots, k$ and $j = 1, \dots, n_i$.

Our objective is to use the differences in the sample means to decide whether any observed differences are due to sampling variation or because the group means are different. We can express this as a hypothesis test, where we have the hypotheses

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$$

$$H_1 : \mu_i \neq \mu_j \text{ for at least one pair}$$



ANOVA

ANOVA -
Analysis of
variance

One-Way ANOVA

Two-Way ANOVA

Matlab and
ANOVA

Recall that the overall mean is denoted by $\bar{x}$. We will represent the mean for the i -th sample as $\bar{x}_i$. The **total sum of squares** measures the dispersion of all the data around the overall mean and is given by

$$\text{Total } SS = \sum_i \sum_j (x_{ij} - \bar{x})^2.$$

We can break this up into two pieces representing variation within groups and between groups, as shown here

$$\sum_i \sum_j (x_{ij} - \bar{x})^2 = \sum_i \sum_j (x_{ij} - \bar{x}_i)^2 + \sum_i (\bar{x}_i - \bar{x})^2 = SSE + SS_T$$

The first term on the right side measures the variation about the mean within groups, and it is called the **error sum of squares** (SSE). The second term is the **treatment sum of squares** SS_T , and it is the variation of the group means with respect to the overall mean $\bar{x}$. The word treatment is used because ANOVA was originally developed for experimental design, where one is interested in comparing the effect of different treatments on an outcome variable.



ANOVA

ANOVA -
Analysis of
variance

One-Way ANOVA

Two-Way ANOVA

Matlab and
ANOVA

We are now ready to develop a test statistic that utilizes the two types of variation on the right-hand side — the SSE and the SS_T . If there is truly no difference in the group means, then it is likely that SSE and SS_T would be approximately equal. If they are very different, then we have evidence against our null hypothesis. The test statistic is the F statistic given by

$$F = \frac{SS_T / (k - 1)}{SSE / (n - k)},$$

where $k - 1$ and $n - k$ are the degrees of freedom.

If the variation within groups (SSE) is small compared to the variation between groups (SS_T), then we would expect to observe a large value of the F statistic.



Assumptions for One-Way ANOVA The standard one-way ANOVA hypothesis tests are valid under the following conditions:

- 1 Independence of observations: the data were collected using statistically valid sampling methods, and there are no hidden relationships among observations. If your data fail to meet this assumption because you have a confounding variable that you need to control for statistically, use an ANOVA with blocking variables.
- 2 Normally-distributed response variable: The values of the dependent variable follow a normal distribution.
- 3 Homogeneity of variance: The variation within each group being compared is similar for every group. If the variances are different among the groups, then ANOVA probably isn't the right fit for the data.



Two-Way ANOVA with interactions

ANOVA -
Analysis of
variance

One-Way ANOVA

Two-Way ANOVA

Matlab and
ANOVA

If we use 2 factors instead of 1 as previously had been shown, the statistical model changes to the following

$$x_{ijk} = \mu + \gamma_i + \beta_j + (\gamma\beta)_{ij} + \epsilon_{ijk},$$

where x_{ijk} represents the k -th measurement made in cell (i,j) of the factorial design, μ is the overall mean, γ_i is the main effect of the first factor, β_j is the main effect of the second factor, $(\gamma\beta)_{ij}$ is the interaction effect of the two factors and ϵ_{ijk} is the residual or error term assumed to have a normal distribution with mean zero and variance σ^2 .

We can leave the $(\gamma\beta)_{ij}$ term from the model, in this case the model is the *2-way ANOVA without interactions*.



Section Overview

ANOVA -
Analysis of
variance

**Matlab and
ANOVA**

One-Way ANOVA

Two-Way ANOVA

Matlab and ANOVA
One-Way ANOVA
Two-Way ANOVA



Matlab and ANOVA

ANOVA -
Analysis of
variance

Matlab and
ANOVA

One-Way ANOVA

Two-Way ANOVA

An anova object contains the results of a one-, two-, or N-way ANOVA. Use the properties of an anova object to determine if the means in a set of response data differ with respect to the values (levels) of a factor or multiple factors. The object properties include information about the coefficient estimates, ANOVA model fit to the response data, and factors used to perform the analysis.



Matlab and ANOVA

ANOVA -
Analysis of
variance

Matlab and
ANOVA

One-Way ANOVA

Two-Way ANOVA

```
aov = anova(y)
```

performs a one-way ANOVA and returns the anova object `aov` for the response data in the matrix `y`. Each column of `y` is treated as a different factor value.

Example Matlab

The columns of the 6-by-3 matrix `popcorn` contain popcorn yield observations in cups for three different brands. Perform a one-way ANOVA to test the null hypothesis that the popcorn yield is not affected by the brand of popcorn.
 H_0 : The brand has no effect on the popcorn yield



Matlab One-Way ANOVA for Matrix Data

ANOVA -
Analysis of
variance

Matlab and
ANOVA

One-Way ANOVA

Two-Way ANOVA

Solution

```
load popcorn.mat
```

```
aov = anova(popcorn)
```

```
aov =
```

```
1-way anova, constrained (Type III) sums of squares.
```

```
Y ~ 1 + Factor1
```

	SumOfSquares	DF	MeanSquares	F	pValue
Factor1	15.75	2	7.875	18.9	7.9603e-05
Error	6.25	15	0.41667		
Total	22	17			

Fig. 2: Result of the One-Way-ANOVA



Matlab One-Way ANOVA for Matrix Data

ANOVA -
Analysis of
variance

Matlab and
ANOVA

One-Way ANOVA

Two-Way ANOVA

- The `SumOfSquares` represents a measure of variation from the mean. It is calculated as a summation of the squares of the differences from the mean.
- `DF` represents the degree of freedom.
- `MeanSquare` is `SumOfSquares` divided by the `DF`.
- The `F`-statistic is the ratio of the mean squares, the `F`-statistic follows an `F`-distribution.
- The `p-value` is calculated using the cumulative distribution function (`cdf`).

The `Factor1` row shows statistics for the model term `Factor1`, and the `Error` row shows statistics for the entire model.



Matlab One-Way ANOVA for Matrix Data

ANOVA -
Analysis of
variance

Matlab and
ANOVA

One-Way ANOVA

Two-Way ANOVA

Conclusion

```
aov =
```

```
1-way anova, constrained (Type III) sums of squares.
```

```
Y ~ 1 + Factor1
```

	SumOfSquares	DF	MeanSquares	F	pValue
	—————	—	—————	———	—————
Factor1	15.75	2	7.875	18.9	7.9603e-05
Error	6.25	15	0.41667		
Total	22	17			

Fig. 3: Result of the One-Way-ANOVA

Conclusion: p -value is small enough that the null hypothesis can be rejected at the 0.01 significance level. Therefore, the brand of popcorn has a significant effect on the popcorn yield.



Matlab Two-Way ANOVA for Vector Data

ANOVA -
Analysis of
variance

Matlab and
ANOVA

One-Way ANOVA

Two-Way ANOVA

Example Matlab

The columns of the 6-by-3 matrix popcorn contain popcorn yield observations in cups for the brands Gourmet, National, and Generic. The first three rows of the matrix correspond to popcorn that was popped with an oil popper, and the last three rows correspond to popcorn that was popped with an air popper.

Create string vectors containing factor values for the brand and popper type. Use the function repmat to repeat copies of strings.

```
brand = [repmat("Gourmet", 6, 1);  
repmat("National", 6, 1); repmat("Generic", 6, 1)];  
poppertype =  
[repmat("Air", 3, 1); repmat("Oil", 3, 1); repmat("Air", 3, 1);  
repmat("Oil", 3, 1); repmat("Air", 3, 1); repmat("Oil", 3, 1)];  
factors = brand, poppertype;
```



Matlab Two-Way ANOVA for Vector Data

ANOVA -
Analysis of
variance

Matlab and
ANOVA

One-Way ANOVA

Two-Way ANOVA

Solution

```
aov = anova(factors, popcorn(:), FactorNames=["Brand"  
"PopperType"])
```

```
aov =  
2-way anova, constrained (Type III) sums of squares.
```

```
Y ~ 1 + Brand + PopperType
```

	SumOfSquares	DF	MeanSquares	F	pValue
Brand	15.75	2	7.875	63	1e-07
PopperType	4.5	1	4.5	36	3.2548e-05
Error	1.75	14	0.125		
Total	22	17			

Fig. 4: Result of the Two-Way-ANOVA



Matlab Two-Way ANOVA for Vector Data

ANOVA -
Analysis of
variance

Matlab and
ANOVA

One-Way ANOVA

Two-Way ANOVA

Conclusion

```
aov =  
2-way anova, constrained (Type III) sums of squares.
```

```
Y ~ 1 + Brand + PopperType
```

	SumOfSquares	DF	MeanSquares	F	pValue
Brand	15.75	2	7.875	63	1e-07
PopperType	4.5	1	4.5	36	3.2548e-05
Error	1.75	14	0.125		
Total	22	17			

Fig. 5: Result of the Two-Way-ANOVA

Conclusion: small p-values indicate that both the brand and popper type have a statistically significant effect on the popcorn yield.



Matlab Two-Way ANOVA for Vector Data

ANOVA -
Analysis of
variance

Matlab and
ANOVA

One-Way ANOVA

Two-Way ANOVA

Further analyzing with the group means: `groupmeans(aov, ["Brand" "PopperType"])`

Brand	PopperType	Mean	SE	MeanLower	MeanUpper
"Gourmet"	"Air"	5.75	0.16667	5.0329	6.4671
"National"	"Air"	4.25	0.16667	3.5329	4.9671
"Generic"	"Air"	3.5	0.16667	2.7829	4.2171
"Gourmet"	"Oil"	6.75	0.16667	6.0329	7.4671
"National"	"Oil"	5.25	0.16667	4.5329	5.9671
"Generic"	"Oil"	4.5	0.16667	3.7829	5.2171

Fig. 6: Group means